Project Title	**IMDB Movie Recommendation System Using Storylines**
Skills take away From This Project	**Selenium, Python, Pandas, Streamlit, NLP (Natural Language Processing), TF-IDF, Cosine Similarity, Machine Learning, Data Cleaning, Data Analysis, Visualization, and User Interface Development.**
Domain	**Entertainment / Data Analytics / Recommender Systems**

Problem Statement:

This project focuses on extracting movie data from IMDb for 2024, specifically focusing on the movie name and storyline. Using **Selenium**, the program will scrape IMDb to collect movie names and their associated storylines. The storylines will then be pre-processed and analyzed using **Natural Language Processing (NLP)** techniques, such as **TF-IDF (Term Frequency-Inverse Document Frequency)** or **Count Vectorizer**. Using these methods, the project will calculate **Cosine Similarity** or other **Machine Learning** algorithms to recommend similar movies based on a given storyline. The project will provide an interactive user interface built with **Streamlit** where users can input a movie storyline and receive the top 5 recommended movies.

Business Use Cases:

1. **Movie Recommendation**: Users can input a movie storyline, and the system will suggest the top 5 most similar movies based on storyline similarity.
2. **Entertainment Suggestions**: Provide personalized movie recommendations based on story preferences.

Approach:

Data Scraping and Storage:

- **Data Source**: IMDb Movies page for 2024.
- **Scraping Method**: Use **Selenium** to extract the following data:
  - Movie Name
  - Storyline (Plot Summary)
- **Data Storage**: Save the extracted movie data into a **CSV file** for further analysis.
  - **Columns**:
    - Movie Name
    - Storyline
- **CSV Storage**: Save the data in a CSV format for easy data manipulation.

Data Preprocessing and Analysis:

- **Text Cleaning (NLP)**:
  - Preprocess the storyline text by removing stop words, punctuation, and unnecessary characters.
  - Tokenize the story text for further analysis.
- **Text Representation**:
  - Use **TF-IDF Vectorizer** or **Count Vectorizer** to convert the movie storylines into numerical vectors.
- **Cosine Similarity**:
  - Calculate the **Cosine Similarity** between the movie storylines to find the most similar movies.
  - Rank the movies based on similarity scores.

Recommendation System:

- **Algorithm**: Use **Cosine Similarity** or other **ML algorithms** to find the top 5 most similar movies based on a user's input.
- **User Input**: Users will be able to input a movie storyline into the **Streamlit app**, and the system will output the top 5 recommended movies based on storyline similarity.

Streamlit Interface:

- **Streamlit Application**:
  - Create an **interactive user interface** using **Streamlit**.
  - Allow users to input a movie storyline (a short description or plot).
  - The app will process the input and display the top 5 recommended movies with their names and storylines.
- **Visualization**:
  - Show the top 5 recommended movies as a **list** with their names and storylines.
  - Provide a **dynamic interface** that allows users to explore different movie recommendations based on storyline input.

Dataset:

- **Scraped IMDb Data for 2024 Movies**.
  - Columns:
    - **Movie Name**: The title of the movie.
    - **Storyline**: The movie's plot summary or storyline.
  - **Format**: CSV file containing the scraped data for all 2024 movies.

Technical Tags:

- **Languages**: Python
- **Libraries/Tools**:
  - **Web Scraping**: Selenium
  - **NLP**: NLTK, SpaCy, Scikit-learn (TF-IDF, Count Vectorizer)
  - **Recommendation Algorithms**: Cosine Similarity, Machine Learning
  - **Web Framework**: Streamlit
  - **Data Manipulation**: Pandas
- **Visualization**: Streamlit, Matplotlib, Seaborn
- **Environment**: VS Code or Python Environment

Project Deliverables:

1. **CSV File**: Containing the scraped IMDb 2024 movie data (Movie Name, Storyline).

2. **Python Scripts**:
    - **Scraping Script**: For extracting the movie data using Selenium.
    - **NLP and Recommendation Script**: For processing the storyline and calculating recommendations using TF-IDF and Cosine Similarity.
    - **Streamlit App Script**: For building the interactive user interface.
3. **Streamlit Application**: A live application where users can input a storyline and receive movie recommendations.
4. **Project Documentation**: A detailed project report explaining the methodology, code, and insights.

Approach Breakdown:

1. **Data Scraping**:
    - Use **Selenium** to scrape IMDb for movie names and storylines from the 2024 movie list.
    - Store the data in a CSV file.
2. **Data Preprocessing**:
    - Clean and tokenize the storylines using **NLP** techniques (removing stop words, punctuation, etc.).
    - Vectorize the text using **TF-IDF Vectorizer** or **Count Vectorizer** to convert text into numerical form.
3. **Cosine Similarity**:
    - Calculate the similarity between each movie's storyline using **Cosine Similarity**.
    - Use the similarity scores to rank movies and make recommendations.
4. **Streamlit Interface**:
    - Create an interactive interface where users can input a movie storyline and receive the top 5 recommended movies based on the cosine similarity score

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Example Use Case:

- **Input**: A user inputs the storyline: “A young wizard begins his journey at a magical school where he makes friends and enemies, facing dark forces along the way.”
- **Output**: The Streamlit app displays the top 5 movies with similar storylines, such as:
    - Movie Name: **The Wizard’s Journey** (Storyline: A boy discovers his magical abilities and faces a powerful sorcerer.)
    - Movie Name: **The Magic Academy** (Storyline: A young student navigates life at a school for magic, facing powerful challenges.)

- Movie Name: **The Dark Sorcerer** (Storyline: A young hero confronts an ancient sorcerer threatening the magical world.)
- And so on, with the most similar movies to the input storyline.

Project Guidelines:

- **Coding Standards**: Follow Python coding conventions and write clean, modular code.
- **Data Validation**: Ensure all data scraped is accurate and complete.
- **Optimized Queries**: Efficient use of Cosine Similarity and other ML algorithms to avoid performance bottlenecks.
- **Documentation**: Provide a well-documented report with a clear explanation of the project and methodology.

Streamlit Doc	https://docs.streamlit.io/library/api-reference
Streamlit recording (Tamil)	**Special Session for STREAMLIT Tamil**
Project Live Evaluation	Project Live Evaluation
GitHub Reference	How to Use GitHub.pptx
Selenium recording	Selenium
Coding Orientation	Coding Guide

Project Evaluation metrics:

- Maintainable: It can be maintained, even as your codebase grows.
- Portable: It works the same in every environment (operating system)
- You have to maintain your code on **GitHub**.(Mandatory)
- You have to keep your **GitHub** repo public so that anyone can check your code.(Mandatory)
- Proper readme file you have to maintain for any project development(Mandatory)
- You should include basic workflow and execution of the entire project in the readme file on **GitHub**
- Follow the coding standards: https://www.python.org/dev/peps/pep-0008/
- You need to Create a Demo video of your working model and post in **LinkedIn**(Mandatory)

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday to Saturday (4:00PM to 5:00PM)

Booking link :https://forms.gle/XC553oSbMJ2Gcfug9

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open on Saturday and Sunday Only on Every Week

Timing: Monday-Saturday (5:30PM to 7:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA